

PRIMARY SCHOOL MANAGEMENT SYSTEM

Technical Design & Implementation Guide

Microsoft Access Database · Biometric-Ready Architecture

Version 1.2 · May 2026

Phase 1: Core Database · Phase 2: Fingerprint Integration

Document Title	Primary School Management System – Tech Guide
Version	1.2 (Corrected & Enhanced)
Date	07 May 2026
Platform	Microsoft Access 2016 / 2019 / Microsoft 365
Database Engine	Microsoft Jet / ACE Engine
Architecture	Split Database (Backend + Frontend)
Status	Phase 1 – Core Implementation (Ready for Build)
Changes from v1.0	See Section 13 – Changelog

Table of Contents

1.	Project Overview	3
2.	System Requirements	4
3.	Database Architecture — Tables & Fields	6
4.	Table Relationships	12
5.	Forms Design Guide	13
6.	Fee Management Module	17
7.	Attendance Module	19
8.	Academic Records Module	21
9.	Reports & Queries	23
10.	VBA Code Snippets (Complete Set)	24
11.	Biometric Integration Plan (Phase 2)	29
12.	Testing Checklist	31
13.	Document Changelog	33

1. Project Overview

The Primary School Management System (PSMS) is a comprehensive desktop database application built on Microsoft Access. It is designed to digitise and streamline every administrative operation of a primary school — from pupil registration and daily attendance tracking, to school fees management and academic result processing. The system is architected in two phases: Phase 1 delivers the full core database with manual attendance, and Phase 2 integrates fingerprint (biometric) verification for pupil dismissal and parent pickup confirmation.

1.1 Objectives

- Maintain a centralised, accurate record of all pupils from KG1 to Primary 5.
- Track daily attendance (arrival and dismissal) with timestamps.
- Verify parent/guardian identity at pickup using biometric thumbprint (Phase 2).
- Manage school fee structures, invoices, payments, and outstanding balances.
- Record and report academic scores per subject, per term, and per session.
- Generate professional reports: report cards, fee receipts, attendance summaries.
- Provide a scalable foundation that grows with the school's pupil population.

1.2 Scope

Module	Phase 1	Phase 2
Pupil Registration	✓ Full	—
Parent / Guardian Records	✓ Full	—
Class & Subject Setup	✓ Full	—
Manual Attendance	✓ Full	—
Biometric Attendance	Placeholder fields	✓ Full
Parent Pickup Biometric	Placeholder fields	✓ Full
Fee Management	✓ Full	—
Academic Records	✓ Full	—
Report Cards	✓ Full	—
Staff Records	✓ Basic	✓ Enhanced

1.3 School Profile Assumptions

- Classes: Kindergarten 1 & 2, Primary 1 through Primary 5 (7 classes total).
- Academic year: 3 terms (First, Second, Third).
- Subjects: English, Mathematics, Basic Science, Social Studies, Civic Education, Religious Studies, Local Language, PHE, Creative Arts.
- Pupil population: Dynamic — new registrations at any time.
- Single-site school (one campus).

■ **Database Size:** Microsoft Access has a hard limit of 2 GB. Monitor actively when pupil count exceeds 1,000. A size-warning function is included in Section 10.8.

2. System Requirements

2.1 Software Requirements

Component	Minimum	Recommended
Operating System	Windows 10	Windows 10/11 (64-bit)
Microsoft Access	Access 2016	Access 2019 / Microsoft 365
Microsoft Office	Office 2016	Office 2019 / 365
VBA Runtime	Included with Access	Included with Access
Biometric SDK (Ph. 2)	Vendor-supplied DLL	Suprema / ZKTeco SDK
Report Viewer	Built-in Access	Built-in Access

2.2 Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i3 or equivalent
RAM	4 GB (8 GB recommended)
Storage	50 GB free disk space
Display	1366 x 768 resolution or higher
Fingerprint Scanner (Ph. 2)	USB biometric reader (ZKTeco, Suprema, DigitalPersona)
Printer	Inkjet or laser for receipts and report cards

2.3 ★ Split Database Architecture (Mandatory)

CRITICAL: For any multi-user deployment you **MUST** split the database. Two users opening the same .accdb file simultaneously **WILL** cause corruption.

Component	Location	File Name
Backend (tables only)	Network shared drive	PSMS_Backend.accdb
Frontend (forms, reports, queries, VBA)	Each user's local PC	PSMS_Frontend.accdb

How to Split:

- Step 1:** Build the complete database with all tables first.
- Step 2:** Go to Database Tools > Access Database > Split Database.
- Step 3:** Distribute frontend copies to each computer in the school.
- Step 4:** Link frontend to backend via External Data > Linked Table Manager.
- Step 5:** Test with two users simultaneously before going live.

2.4 Access Database Settings

- Set Access Trust Center to allow macros from the database location.
- Enable VBA macro execution: Access Options > Trust Center > Macro Settings.
- Save as .accdb format (Access 2007–365).
- Enable Microsoft ActiveX Data Objects (ADO) library in VBA Tools > References.
- Set startup form to frm_MainDashboard.
- Password-protect: File > Info > Encrypt with Password.

2.5 ★ Photo Storage — Important

DO NOT store photos as OLE Objects. OLE embeds images inside the .accdb file and will bloat your database to the 2 GB limit very quickly.

CORRECT APPROACH: Store only the file path in a Short Text field (255 chars). Keep all images in a dedicated folder: C:\SchoolData\Photos

- Field in tbl_Pupils: **PhotoPath** (Short Text, 255) — NOT Photo (OLE Object).
- Example value: C:\SchoolData\Photos\2024-0001.jpg
- Use a VBA file picker to let staff browse and select the photo file.
- Display using an Image control bound to the PhotoPath field.
- Back up the Photos folder separately from the database file.

3. Database Architecture — Tables & Fields

The PSMS database consists of 13 core tables organised into three categories: Reference/Setup tables, Core Entity tables, and Transaction tables. Every table uses an AutoNumber primary key.

3.1 tbl_Classes

Stores the list of classes. Populate this table first before any other data entry.

Field Name	Data Type	Size	Required	Description
ClassID	AutoNumber	Long Int	PK	Unique class identifier
ClassName	Short Text	50	Yes	KG1, KG2, Primary 1 ... Primary 5
ClassLevel	Number	Integer	Yes	Order: 1=KG1 ... 7=Primary 5
ClassTeacherID	Number	Long Int	No	FK → tbl_Staff.StaffID
Capacity	Number	Integer	No	Maximum pupil capacity
IsActive	Yes/No	—	Yes	Deactivate without deleting

3.2 tbl_Subjects

Field Name	Data Type	Size	Required	Description
SubjectID	AutoNumber	Long Int	PK	Unique subject identifier
SubjectName	Short Text	100	Yes	Full name e.g. English Language
SubjectCode	Short Text	10	Yes	Short code: ENG, MTH, BSC...
IsActive	Yes/No	—	Yes	Enable / disable subject

3.3 tbl_Pupils ■ Core Table

Photo is stored as a FILE PATH (PhotoPath), NOT as an OLE Object. ThumbprintData is Null in Phase 1 — the field is reserved for Phase 2.

Field Name	Data Type	Size	Required	Description
PupilID	AutoNumber	Long Int	PK	Unique pupil identifier
AdmissionNo	Short Text	20	Yes	Unique admission number (indexed, no dupes)
FirstName	Short Text	50	Yes	Pupil's first name
LastName	Short Text	50	Yes	Pupil's surname
MiddleName	Short Text	50	No	Middle name (optional)
Gender	Short Text	10	Yes	Male / Female
DateOfBirth	Date/Time	—	Yes	Format: DD/MM/YYYY

ClassID	Number	Long Int	Yes	FK → tbl_Classes.ClassID
AdmissionDate	Date/Time	—	Yes	Date admitted to school
StatusID	Number	Long Int	Yes	FK → tbl_PupilStatus.StatusID
BloodGroup	Short Text	5	No	A+, B-, O+, etc.
Allergies	Long Text	—	No	Known allergies
MedicalNotes	Long Text	—	No	Other medical information
PhotoPath	Short Text	255	No	File path to photo — NOT OLE Object
ThumbprintData	OLE Object	—	No	Biometric template (Phase 2 — leave Null)
ThumbprintEnrolled	Yes/No	—	Yes	Has fingerprint been captured?
DateCreated	Date/Time	—	Yes	Auto-set on new record
LastModified	Date/Time	—	Yes	Auto-updated on edit

■ *Note: Set AdmissionNo as Indexed (No Duplicates) in the field properties to prevent duplicate admission numbers at the database level.*

3.4 tbl_Parents_Guardians

Stores parent and guardian details. Multiple guardians can link to one pupil through tbl_PupilGuardian (Section 3.5).

Field Name	Data Type	Size	Required	Description
ParentID	AutoNumber	Long Int	PK	Unique parent/guardian identifier
Title	Short Text	10	No	Mr, Mrs, Dr, Prof, etc.
FirstName	Short Text	50	Yes	First name
LastName	Short Text	50	Yes	Surname
Relationship	Short Text	20	Yes	Father / Mother / Guardian / Uncle...
PhoneNumber1	Short Text	15	Yes	Primary contact number
PhoneNumber2	Short Text	15	No	Secondary number
Email	Short Text	100	No	Email address
HomeAddress	Long Text	—	No	Residential address
Occupation	Short Text	100	No	Occupation
NationalID	Short Text	30	No	National ID or passport number
ThumbprintData	OLE Object	—	No	Biometric template (Phase 2)
ThumbprintEnrolled	Yes/No	—	Yes	Fingerprint captured?
IsAuthorizedPickup	Yes/No	—	Yes	Allowed to pick up child?
DateCreated	Date/Time	—	Yes	Record creation timestamp

3.5 tbl_PupilGuardian (Junction Table)

Links pupils to their parents/guardians. One pupil can have multiple guardians.

Field Name	Data Type	Size	Required	Description
PupilGuardianID	AutoNumber	Long Int	PK	Auto ID
PupilID	Number	Long Int	Yes	FK → tbl_Pupils.PupilID
ParentID	Number	Long Int	Yes	FK → tbl_Parents_Guardians.ParentID
IsPrimaryContact	Yes/No	—	Yes	Primary emergency contact?
CanPickUp	Yes/No	—	Yes	Authorised for daily pickup?

3.6 tbl_Attendance

Records daily attendance. Phase 1: manual entry. Phase 2: biometric auto-populate. One record per pupil per date — enforced by the save logic in Section 7.2.

Field Name	Data Type	Size	Required	Description
------------	-----------	------	----------	-------------

AttendanceID	AutoNumber	Long Int	PK	Auto ID
PupilID	Number	Long Int	Yes	FK → tbl_Pupils.PupilID
AttendanceDate	Date/Time	—	Yes	Date of attendance record
ArrivalTime	Date/Time	—	No	Time pupil arrived
DismissalTime	Date/Time	—	No	Time pupil was dismissed
Status	Short Text	15	Yes	Present / Absent / Late / Excused
ArrivalMethod	Short Text	15	Yes	Manual / Biometric / ManualOverride
MarkedByStaffID	Number	Long Int	No	FK → tbl_Staff.StaffID
Remarks	Short Text	200	No	Additional notes

3.7 tbl_PickupLog

Safety-critical table. Every parent pickup is logged with timestamp and method.

Field Name	Data Type	Size	Required	Description
PickupID	AutoNumber	Long Int	PK	Auto ID
PupilID	Number	Long Int	Yes	FK → tbl_Pupils.PupilID
ParentID	Number	Long Int	Yes	FK → tbl_Parents_Guardians.ParentID
PickupDateTime	Date/Time	—	Yes	Date and time of pickup
VerificationMethod	Short Text	15	Yes	Manual / Biometric / ManualOverride
BiometricMatch	Yes/No	—	No	Fingerprint verified? (Phase 2)
VerifiedByStaffID	Number	Long Int	No	Staff who confirmed pickup
OverridePINUsed	Yes/No	—	Yes	Emergency override PIN was used?
Notes	Short Text	200	No	Remarks

3.8 tbl_FeeStructure

Field Name	Data Type	Size	Required	Description
FeeStructureID	AutoNumber	Long Int	PK	Auto ID
ClassID	Number	Long Int	Yes	FK → tbl_Classes.ClassID
FeeName	Short Text	100	Yes	Tuition, Books, Uniform, PTA...
Amount	Currency	—	Yes	Amount in Naira
TermNumber	Number	Integer	Yes	1, 2, or 3
AcademicYearID	Number	Long Int	Yes	FK → tbl_AcademicYears.YearID
IsMandatory	Yes/No	—	Yes	Compulsory fee?
DueDate	Date/Time	—	No	Payment deadline

■ **NEW (v1.2):** AcademicYearID now links to tbl_AcademicYears instead of a free-text field. This prevents typo mismatches like '2024/25' vs '2024/2025'.

3.9 tbl_FeePayments

Field Name	Data Type	Size	Required	Description
PaymentID	AutoNumber	Long Int	PK	Auto ID
PupilID	Number	Long Int	Yes	FK → tbl_Pupils.PupilID
FeeStructureID	Number	Long Int	Yes	FK → tbl_FeeStructure.FeeStructureID
AmountPaid	Currency	—	Yes	Amount paid this transaction
PaymentDate	Date/Time	—	Yes	Date payment received

PaymentMethod	Short Text	20	Yes	Cash / Bank Transfer / POS
ReceiptNumber	Short Text	25	Yes	Auto-generated: RCP-YYYY-NNNNN
ReceivedByStaffID	Number	Long Int	Yes	FK → tbl_Staff.StaffID
BankReference	Short Text	50	No	Bank teller / transfer reference
Notes	Short Text	200	No	Additional notes

■ **NEW (v1.2):** ReceiptNumber format is now RCP-YYYY-NNNNN (year included) for easier annual auditing.

3.10 tbl_ExamScores

CA = Continuous Assessment (40 marks total). Exam = 60 marks. Total = 100.

Field Name	Data Type	Size	Required	Description
ScoreID	AutoNumber	Long Int	PK	Auto ID
PupilID	Number	Long Int	Yes	FK → tbl_Pupils.PupilID
SubjectID	Number	Long Int	Yes	FK → tbl_Subjects.SubjectID
ClassID	Number	Long Int	Yes	FK → tbl_Classes.ClassID
TermNumber	Number	Integer	Yes	1, 2, or 3
AcademicYearID	Number	Long Int	Yes	FK → tbl_AcademicYears.YearID
CA1Score	Number	Single	No	First CA (max 20)
CA2Score	Number	Single	No	Second CA (max 20)
ExamScore	Number	Single	No	Exam score (max 60)
TotalScore	Number	Single	No	CA1+CA2+Exam (calculated field)
Grade	Short Text	5	No	A–F (auto from GradeFromScore())
TeacherRemark	Short Text	100	No	Teacher's comment
EnteredByStaffID	Number	Long Int	No	FK → tbl_Staff.StaffID

3.11 tbl_Staff

Field Name	Data Type	Size	Required	Description
StaffID	AutoNumber	Long Int	PK	Unique staff identifier
StaffNo	Short Text	20	Yes	Employee number
FirstName	Short Text	50	Yes	First name
LastName	Short Text	50	Yes	Surname
Role	Short Text	50	Yes	Teacher / Admin / Principal / Security
PhoneNumber	Short Text	15	Yes	Contact number
Email	Short Text	100	No	Work email
IsActive	Yes/No	—	Yes	Currently employed?
Username	Short Text	30	No	Login username (future use)
PasswordHash	Short Text	100	No	Hashed password (future use)

3.12 tbl_PupilStatus (Lookup Table)

Field Name	Data Type	Size	Required	Description
------------	-----------	------	----------	-------------

StatusID	AutoNumber	Long Int	PK	Auto ID
StatusName	Short Text	20	Yes	Active / Left / Graduated / Transferred

Pre-populate this table immediately after creating it with these 4 records: 1=Active, 2=Left, 3=Graduated, 4=Transferred. Use StatusID=1 (Active) when filtering for current pupils in queries and reports.

3.13 tbl_AcademicYears ■ (Now Mandatory — v1.2)

■ **NEW (v1.2):** Upgraded from Optional to Mandatory. Prevents typo mismatches in academic year fields across FeeStructure and ExamScores tables.

Field Name	Data Type	Size	Required	Description
YearID	AutoNumber	Long Int	PK	Auto ID
YearName	Short Text	15	Yes	e.g. 2024/2025
StartDate	Date/Time	—	Yes	First day of academic year
EndDate	Date/Time	—	Yes	Last day of academic year
IsCurrent	Yes/No	—	Yes	Only one record should be Yes at a time

■ **Note:** All dropdowns for Academic Year across all forms should be sourced from this table (YearName field), not free-text entry. This ensures consistency across all queries and reports.

4. Table Relationships

Build all relationships in Database Tools > Relationships. Enable Referential Integrity and Cascade Update Related Fields for every relationship. Do NOT enable Cascade Delete unless you are absolutely certain — it can silently remove child records.

Parent Table	Child Table	Join Field	Type
tbl_Classes	tbl_Pupils	ClassID	One-to-Many
tbl_Pupils	tbl_PupilGuardian	PupilID	One-to-Many
tbl_Parents_Guardians	tbl_PupilGuardian	ParentID	One-to-Many
tbl_Pupils	tbl_Attendance	PupilID	One-to-Many
tbl_Pupils	tbl_PickupLog	PupilID	One-to-Many
tbl_Parents_Guardians	tbl_PickupLog	ParentID	One-to-Many
tbl_Classes	tbl_FeeStructure	ClassID	One-to-Many
tbl_Pupils	tbl_FeePayments	PupilID	One-to-Many
tbl_FeeStructure	tbl_FeePayments	FeeStructureID	One-to-Many
tbl_Pupils	tbl_ExamScores	PupilID	One-to-Many
tbl_Subjects	tbl_ExamScores	SubjectID	One-to-Many
tbl_Classes	tbl_ExamScores	ClassID	One-to-Many
tbl_Staff	tbl_Classes	ClassTeacherID	One-to-Many
tbl_Staff	tbl_Attendance	MarkedByStaffID	One-to-Many
tbl_Staff	tbl_FeePayments	ReceivedByStaffID	One-to-Many
tbl_PupilStatus	tbl_Pupils	StatusID	One-to-Many
tbl_AcademicYears	tbl_FeeStructure	AcademicYearID	One-to-Many
tbl_AcademicYears	tbl_ExamScores	AcademicYearID	One-to-Many

Total: 18 relationships. The last 3 (tbl_PupilStatus and tbl_AcademicYears) are new in v1.2. Build all 18 before creating any forms.

5. Forms Design Guide

Build all forms in Form Design view. Colour scheme: Navy (#1B3A6B) header, white body, light blue (#DBEAFE) subforms. Use tab order and keyboard shortcuts to make data entry fast for busy school staff.

5.1 frm_MainDashboard (Startup Form)

- Header: School logo (image file — not OLE), school name label, current date/time (auto-updating).
- Navigation buttons: Pupils, Parents, Attendance, Dismissal, Pickup Log, Fees, Exam Scores, Reports, Settings.
- Quick stats: Total Active Pupils (DCount), Today's Attendance %, Fee Defaulters count.
- Database size indicator using CheckDatabaseSize() function (Section 10.8).
- Footer: Logged-in user name, version number.

5.2 frm_PupilRegistration

- AdmissionNo auto-generated via Form_BeforeInsert event (Section 10.1).
- Tab Control — Tab 1 Personal: FirstName, LastName, MiddleName, Gender, DOB, Class, AdmissionDate.
- Tab Control — Tab 2 Medical: BloodGroup, Allergies, MedicalNotes.
- Tab Control — Tab 3 Guardians: subform frm_PupilGuardian_Sub with CanPickUp checkbox.
- Photo display: Image control bound to PhotoPath. 'Pick Photo' button calls PickPhoto() (Section 10.7).
- StatusID combo box sourced from tbl_PupilStatus.
- Buttons: Save, Add New, Delete, Print Admission Slip, Search Pupil.
- Disabled placeholder: 'Enrol Thumbprint — Coming in Phase 2'.

5.3 frm_ParentGuardian

- Fields: Title, FirstName, LastName, Relationship, Phone1, Phone2, Email, Address, Occupation, NationalID.
- IsAuthorizedPickup checkbox — large and prominent (safety-critical field).
- Linked Pupils subform: all children linked via tbl_PupilGuardian.
- Buttons: Save, New, Delete, Print Parent ID Card.
- Disabled placeholder: 'Enrol Thumbprint — Coming in Phase 2'.

5.4 frm_Attendance (Morning Arrival)

- Filters: Class combo, Date field (defaults to today via =Date()).
- Continuous subform: PupilName, AdmissionNo, Status combo, ArrivalTime, Remarks.
- 'Mark All Present' button — sets all to Present with Now() timestamp (Section 10.4).
- 'Save Attendance' uses INSERT/UPDATE logic — never creates duplicate records (Section 7.2).
- Summary counts at bottom: Present, Absent, Late (use DCount queries).

CRITICAL: The Save button must check if a record already exists for PupilID + AttendanceDate before deciding to INSERT or UPDATE. Never use a simple append query here.

5.5 ★ frm_Dismissal (New — End of Day)

■ **NEW (v1.1): This form was missing in v1.0. It is required for a complete workflow.**

- Filters: Class combo, Date field (defaults to today).
- Continuous subform: PupilName, AdmissionNo, ArrivalTime (read-only), DismissalTime (editable), Remarks.
- 'Mark All Dismissed' button — sets DismissalTime = Now() for all Present pupils.
- 'Save Dismissals' button — UPDATES existing tbl_Attendance records.
- Summary: Present count, Dismissed count, Still on Campus count.

5.6 frm_PickupLog (Safety-Critical)

- Search pupil by AdmissionNo or name — shows photo (from PhotoPath), name, class.
- Authorised Guardians list: sourced from tbl_PupilGuardian WHERE CanPickUp = Yes.
- Staff selects guardian → confirmation dialog → inserts record into tbl_PickupLog.
- Pickup timestamp shown immediately on confirmation.
- Emergency override PIN field (admin-only) for non-registered pickups — logs OverridePINUsed = Yes.
- Phase 2 placeholder: disabled 'Verify Thumbprint' button.

5.7 frm_FeeStructure (Admin Only)

- Select Class, Term, Academic Year (from tbl_AcademicYears dropdown).
- Enter fee items and amounts.
- 'Copy from Previous Term' button — saves setup time for recurring terms.

5.8 frm_FeeInvoice

- Select pupil → system auto-loads class fee structure.
- Subform shows: Fee Item | Amount Due | Amount Paid | Balance.
- Total Due, Total Paid, Outstanding Balance from qry_FeeSummary.
- Print Invoice button opens rpt_FeeInvoice report.

5.9 frm_FeePayment

- Search pupil by AdmissionNo or name.
- FeeStructureID combo shows fee items for pupil's class and selected term.
- AmountPaid validated against outstanding balance (Section 10.5).
- ReceiptNumber auto-generated: RCP-YYYY-NNNNN (Section 10.2).
- PaymentMethod combo: Cash / Bank Transfer / POS.
- Print Receipt button opens rpt_FeeReceipt.

5.10 frm_ExamScores

- Filters: Class, Subject, Term, Academic Year (from tbl_AcademicYears).
- Continuous subform: PupilName, CA1Score, CA2Score, ExamScore.
- TotalScore calculated: =[CA1Score]+[CA2Score]+[ExamScore].
- Grade auto-assigned: =GradeFromScore([TotalScore]).

- Tab order optimised for fast bulk entry across columns.

6. Fee Management Module

6.1 ★ Corrected Fee Summary Query (qry_FeeSummary)

■ **FIXED (v1.1):** v1.0 used **INNER JOIN** which excluded pupils with no payment records. Corrected to **LEFT JOIN** so all pupils appear even with zero payments.

```
SELECT
    p.AdmissionNo,
    p.FirstName & " " & p.LastName AS PupilName,
    c.ClassName,
    fs.FeeName,
    fs.Amount AS AmountDue,
    Nz(SUM(fp.AmountPaid), 0) AS TotalPaid,
    fs.Amount - Nz(SUM(fp.AmountPaid), 0) AS Balance
FROM
    (tbl_Pupils p
    INNER JOIN tbl_Classes c ON p.ClassID = c.ClassID)
    LEFT JOIN tbl_FeeStructure fs
        ON c.ClassID = fs.ClassID AND fs.AcademicYearID = [Enter Year ID:]
    LEFT JOIN tbl_FeePayments fp
        ON (p.PupilID = fp.PupilID AND fs.FeeStructureID = fp.FeeStructureID)
WHERE
    fs.TermNumber = [Enter Term (1,2,3):]
    AND p.StatusID = 1
GROUP BY
    p.AdmissionNo, p.FirstName, p.LastName,
    c.ClassName, fs.FeeName, fs.Amount
ORDER BY
    c.ClassName, p.LastName;
```

6.2 Fee Workflow Summary

Step	Form / Report	Notes
Admin sets up fee structure	frm_FeeStructure	Once per term, before any payments
Bursar opens pupil invoice	frm_FeeInvoice	Shows what is owed per fee item
Parent makes payment	frm_FeePayment	Amount validated, receipt generated
Receipt printed	rpt_FeeReceipt	Given to parent as proof of payment
Admin reviews defaulters	rpt_FeeDefaulters	Pupils with outstanding balance > 0

7. Attendance Module

7.1 Daily Workflow

Step	Action	Form
1	Teacher selects class and today's date	frm_Attendance
2	System loads all active pupils for that class	(automatic)
3	Teacher marks: Present / Absent / Late / Excused	frm_Attendance
4	ArrivalTime auto-stamped for Present and Late pupils	(automatic)
5	Teacher clicks Save — INSERT or UPDATE (no duplicates)	frm_Attendance
6	End of day: staff opens Dismissal form	frm_Dismissal
7	Staff records DismissalTime for each Present pupil	frm_Dismissal
8	Parent arrives — pickup confirmed and logged	frm_PickupLog

7.2 ★ Save Attendance — INSERT / UPDATE Logic

■ **FIXED (v1.1): v1.0 used a simple append which caused duplicate records. The corrected code below checks for existing records before deciding INSERT or UPDATE.**

```
Private Sub btnSaveAttendance_Click()  
    Dim rs As DAO.Recordset  
    Dim pupilID As Long  
    Dim statusValue As String  
    Dim arrivalVal As Variant  
  
    Set rs = Me.frm_AttendanceSub.Form.RecordsetClone  
    If rs.RecordCount = 0 Then Exit Sub  
    rs.MoveFirst  
  
    Do While Not rs.EOF  
        pupilID = rs!PupilID  
        statusValue = Nz(rs!Status, "Absent")  
  
        If statusValue = "Present" Or statusValue = "Late" Then  
            arrivalVal = Nz(rs!ArrivalTime, Now())  
        Else  
            arrivalVal = Null  
        End If  
  
        If DCount("*", "tbl_Attendance", "PupilID=" & pupilID & _  
            " AND AttendanceDate=#" & Format(Me.txtDate,"yyyy/mm/dd") & "#") > 0 Then  
            ' -- UPDATE existing record --  
            CurrentDb.Execute "UPDATE tbl_Attendance SET " & _  
                "Status='" & statusValue & "', " & _  
                "ArrivalTime=" & IIf(IsNull(arrivalVal),"NULL","#") & arrivalVal & "#") & ", " &  
            _  
                "Remarks='" & Nz(rs!Remarks,"") & "' " & _  
                "WHERE PupilID=" & pupilID & _  
                " AND AttendanceDate=#" & Format(Me.txtDate,"yyyy/mm/dd") & "#"  
        Else  
            ' -- INSERT new record --  
            CurrentDb.Execute "INSERT INTO tbl_Attendance (PupilID, Status, ArrivalTime, Remarks) VALUES (" & pupilID & ", '" & statusValue & "', " & _  
                "ArrivalTime=" & IIf(IsNull(arrivalVal),"NULL","#") & arrivalVal & "#', '" & Nz(rs!Remarks,"") & "')" & _  
            _  
                " AND AttendanceDate=#" & Format(Me.txtDate,"yyyy/mm/dd") & "#")  
        End If  
        rs.MoveNext  
    End Do  
End Sub
```

```

Else
    ' -- INSERT new record --
    CurrentDb.Execute "INSERT INTO tbl_Attendance " & _
        "(PupilID,AttendanceDate,Status,ArrivalTime,Remarks) VALUES (" & _
        pupilID & ",#" & Format(Me.txtDate,"yyyy/mm/dd") & "#,'" & _
        statusValue & "','" & _
        IIf(IsNull(arrivalVal),"NULL","#" & arrivalVal & "#") & _
        "','" & Nz(rs!Remarks,"") & "'"")
End If
rs.MoveNext
Loop

MsgBox "Attendance saved successfully.", vbInformation
End Sub

```

7.3 Attendance Summary Query

```

SELECT
    p.FirstName & " " & p.LastName AS PupilName,
    c.ClassName,
    COUNT(a.AttendanceID) AS TotalDays,
    SUM(IIf(a.Status='Present',1,0)) AS DaysPresent,
    SUM(IIf(a.Status='Absent', 1,0)) AS DaysAbsent,
    SUM(IIf(a.Status='Late', 1,0)) AS DaysLate,
    Format(SUM(IIf(a.Status='Present',1,0))/COUNT(a.AttendanceID)*100,"0.0") & "%" AS Rate
FROM tbl_Pupils p
    INNER JOIN tbl_Classes c ON p.ClassID = c.ClassID
    INNER JOIN tbl_Attendance a ON p.PupilID = a.PupilID
WHERE a.AttendanceDate BETWEEN [Start Date:] AND [End Date:]
    AND p.StatusID = 1
GROUP BY p.FirstName, p.LastName, c.ClassName
ORDER BY c.ClassName, p.LastName;

```

8. Academic Records Module

8.1 Grading Scale

Score Range	Grade	Remark
70 – 100	A	Excellent
60 – 69	B	Very Good
50 – 59	C	Good
40 – 49	D	Pass
0 – 39	F	Fail

8.2 Grade Calculation Function

```
Function GradeFromScore(score As Variant) As String
    If IsNull(score) Then GradeFromScore = "": Exit Function
    Select Case CDb1(score)
        Case 70 To 100: GradeFromScore = "A"
        Case 60 To 69:  GradeFromScore = "B"
        Case 50 To 59:  GradeFromScore = "C"
        Case 40 To 49:  GradeFromScore = "D"
        Case Else:      GradeFromScore = "F"
    End Select
End Function
```

8.3 Class Position Function

■ *Note: For classes under 200 pupils use the DCount method below. For larger classes consider a temporary table or archive old academic years.*

```
Function GetClassPosition(pupilID As Long, classID As Long, _
    termNum As Integer, yearID As Long) As Integer
    Dim pupilTotal As Double
    Dim higherCount As Integer

    pupilTotal = Nz(DSum("CA1Score+CA2Score+ExamScore", "tbl_ExamScores", _
        "PupilID=" & pupilID & " AND ClassID=" & classID & _
        " AND TermNumber=" & termNum & " AND AcademicYearID=" & yearID), 0)

    higherCount = DCount("qry_PupilGrandTotals", _
        "GrandTotal>" & pupilTotal & _
        " AND ClassID=" & classID & _
        " AND TermNumber=" & termNum & _
        " AND AcademicYearID=" & yearID)

    GetClassPosition = higherCount + 1
End Function
```

■ *Note: qry_PupilGrandTotals must be a saved query that calculates SUM(CA1+CA2+Exam) per pupil per class per term — used as the basis for ranking.*

8.4 Result Summary Query

```

SELECT
    p.AdmissionNo,
    p.FirstName & " " & p.LastName AS PupilName,
    s.SubjectName, s.SubjectCode,
    e.CA1Score, e.CA2Score, e.ExamScore,
    e.CA1Score + e.CA2Score + e.ExamScore AS Total,
    e.Grade, e.TeacherRemark
FROM tbl_Pupils p
    INNER JOIN tbl_ExamScores e ON p.PupilID = e.PupilID
    INNER JOIN tbl_Subjects s    ON e.SubjectID = s.SubjectID
WHERE e.ClassID = [ClassID]
    AND e.TermNumber = [TermNumber]
    AND e.AcademicYearID = [AcademicYearID]
ORDER BY p.LastName, s.SubjectName;

```

9. Reports & Queries

Report Name	Purpose	Key Data Source
rpt_ReportCard	Academic result slip per pupil per term	qry_ResultSummary
rpt_AttendanceRegister	Daily register by class	tbl_Attendance
rpt_AttendanceSummary	Percentage attendance, flags below 75%	qry_AttendanceSummary
rpt_FeeReceipt	Official payment receipt	tbl_FeePayments
rpt_FeeInvoice	Per-pupil invoice showing amounts due	qry_FeeSummary
rpt_FeeDefaulters	Pupils with outstanding balance > 0	qry_FeeSummary
rpt_FeeCollectionSummary	Total collected vs expected per class/term	qry_FeeSummary
rpt_PupilList	Active pupils with contact details	tbl_Pupils
rpt_PickupLog	Daily pickup audit trail	tbl_PickupLog
rpt_ClassPerformance	Average score per subject per class	tbl_ExamScores
rpt_EmergencyContacts	Class list with parent contacts (printable)	qry_EmergencyContacts

9.1 ★ rpt_EmergencyContacts Query (New)

■ **NEW (v1.1): Added in v1.1. Essential operational report for every school.**

```

SELECT
    c.ClassName,
    p.FirstName & " " & p.LastName AS PupilName,
    p.AdmissionNo,
    pg.FirstName & " " & pg.LastName AS ParentName,
    pg.Relationship,
    pg.PhoneNumber1,
    pg.PhoneNumber2,
    pj.IsPrimaryContact,
    pj.CanPickUp
FROM ((tbl_Classes c
      INNER JOIN tbl_Pupils p    ON c.ClassID = p.ClassID)
      INNER JOIN tbl_PupilGuardian pj ON p.PupilID = pj.PupilID)
      INNER JOIN tbl_Parents_Guardians pg ON pj.ParentID = pg.ParentID
WHERE pj.IsPrimaryContact = True
      AND p.StatusID = 1
ORDER BY c.ClassName, p.LastName;
```


10. VBA Code Snippets — Complete Set

Place all module-level functions (10.1–10.3, 10.6–10.8) in a Standard Module (Insert > Module) named modUtilities. Place form-specific event handlers on each form's code behind.

10.1 Auto-Generate Admission Number

```
Function GenerateAdmissionNo() As String
    Dim lastNo As Long
    Dim yr As String
    yr = Format(Year(Date), "0000")
    lastNo = Nz(DMax("Val(Mid(AdmissionNo,6))", "tbl_Pupils", _
        "Left(AdmissionNo,4)=" & Chr(34) & yr & Chr(34)), 0)
    GenerateAdmissionNo = yr & "-" & Format(lastNo + 1, "0000")
End Function

Private Sub Form_BeforeInsert(Cancel As Integer)
    Me.AdmissionNo = GenerateAdmissionNo()
End Sub
```

10.2 ★ Auto-Generate Receipt Number (Year-Inclusive Format)

■ NEW (v1.2): Format updated to RCP-YYYY-NNNNN to include the year for audit trails.

```
Function GenerateReceiptNo() As String
    Dim lastNo As Long
    Dim yr As String
    yr = Format(Year(Date), "0000")
    lastNo = Nz(DMax("Val(Mid(ReceiptNumber,10))", "tbl_FeePayments", _
        "Left(ReceiptNumber,8)='RCP-" & yr & "'"), 0)
    GenerateReceiptNo = "RCP-" & yr & "-" & Format(lastNo + 1, "00000")
End Function
' Example output: RCP-2025-00001
```

10.3 Grade Calculation Function

```
Function GradeFromScore(score As Variant) As String
    If IsNull(score) Then GradeFromScore = "": Exit Function
    Select Case CDBl(score)
        Case 70 To 100: GradeFromScore = "A"
        Case 60 To 69:  GradeFromScore = "B"
        Case 50 To 59:  GradeFromScore = "C"
        Case 40 To 49:  GradeFromScore = "D"
        Case Else:      GradeFromScore = "F"
    End Select
End Function
```

10.4 Mark All Present Button

```
Private Sub btnMarkAllPresent_Click()
    If MsgBox("Mark all pupils Present?", vbYesNo + vbQuestion) = vbNo Then Exit Sub

    Dim rs As DAO.Recordset
    Set rs = Me.frm_AttendanceSub.Form.RecordsetClone
    If rs.RecordCount = 0 Then Exit Sub
```

```

rs.MoveFirst
Do While Not rs.EOF
    rs.Edit
    rs!Status      = "Present"
    rs!ArrivalTime = Now()
    rs.Update
    rs.MoveNext
Loop

Me.frm_AttendanceSub.Form.Requery
MsgBox "All pupils marked Present.", vbInformation
End Sub

```

10.5 Fee Overpayment Validation

```

Private Sub AmountPaid_BeforeUpdate(Cancel As Integer)
    Dim amountDue As Currency
    Dim totalPaid As Currency
    Dim balance As Currency

    amountDue = Nz(DLookup("Amount", "tbl_FeeStructure", _
        "FeeStructureID=" & Me.FeeStructureID), 0)
    totalPaid = Nz(DSum("AmountPaid", "tbl_FeePayments", _
        "FeeStructureID=" & Me.FeeStructureID & " AND PupilID=" & Me.PupilID), 0)
    balance = amountDue - totalPaid

    If Me.AmountPaid > balance Then
        MsgBox "Amount exceeds outstanding balance of N" & _
            Format(balance, "#,##0.00"), vbExclamation, "Overpayment"
        Cancel = True
    End If
End Sub

```

10.6 Pupil Search

```

Private Sub btnSearch_Click()
    Dim searchTerm As String
    searchTerm = InputBox("Enter Admission No or Pupil Name:", "Search Pupil")
    If searchTerm = "" Then Exit Sub

    Me.Filter = "AdmissionNo Like '*' & searchTerm & '*' OR " & _
        "FirstName & ' ' & LastName Like '*' & searchTerm & '*'"
    Me.FilterOn = True
End Sub

```

10.7 Photo File Picker (Replaces OLE)

■ **NEW (v1.1): Added in v1.1 — replaces OLE Object approach to prevent database bloat.**

```

Private Function PickPhoto() As String
    Dim fd As FileDialog
    Set fd = Application.FileDialog(msoFileDialogFilePicker)

    With fd
        .Title = "Select Pupil Photo"
        .Filters.Clear
        .Filters.Add "Images", "*.jpg;*.jpeg;*.png;*.bmp"
    End With

```

```

        .AllowMultiSelect = False

    If .Show = -1 Then
        ' Optional: copy file to school photos folder
        ' FileCopy .SelectedItems(1), "C:\SchoolData\Photos\" & Me.AdmissionNo & ".jpg"
        PickPhoto = .SelectedItems(1)
    Else
        PickPhoto = ""
    End If
End With
Set fd = Nothing
End Function

Private Sub btnPickPhoto_Click()
    Dim path As String
    path = PickPhoto()
    If path <> "" Then
        Me.PhotoPath = path
        Me.imgPhoto.Picture = path ' bind to Image control
    End If
End Sub

```

10.8 Database Size Warning

■ **NEW (v1.1):** Added in v1.1 — call from frm_MainDashboard On_Open event.

```

Function CheckDatabaseSize() As Boolean
    Dim dbSizeMB As Double
    dbSizeMB = FileLen(CurrentDb.Name) / 1048576 ' bytes to MB

    If dbSizeMB > 1800 Then
        MsgBox "WARNING: Database is " & Format(dbSizeMB,"0") & " MB." & vbCrLf & _
            "Access limit is 2048 MB." & vbCrLf & _
            "Please archive old data or compact the database immediately.", _
            vbExclamation, "Storage Warning"
        CheckDatabaseSize = True
    ElseIf dbSizeMB > 1500 Then
        MsgBox "Note: Database is " & Format(dbSizeMB,"0") & " MB. Monitor closely.", _
            vbInformation, "Storage Notice"
        CheckDatabaseSize = False
    Else
        CheckDatabaseSize = False
    End If
End Function

Private Sub Form_Open(Cancel As Integer)
    CheckDatabaseSize
End Sub

```

11. Biometric Integration Plan — Phase 2

11.1 Recommended Hardware (Nigerian Market Prices)

Brand	Model	Interface	SDK Language	Approx. Cost (₦)
ZKTeco	ZK4500 / SilkFP-100H	USB	VB.NET / C# DLL	₦15,000 – ₦30,000
Suprema	BioMini Slim 2	USB	C++ / .NET SDK	₦25,000 – ₦50,000
DigitalPersona	U.are.U 4500	USB	VB6 / .NET SDK	₦20,000 – ₦40,000
Nitgen	Hamster III	USB	ActiveX / .NET	₦18,000 – ₦35,000

11.2 ★ Recommended Integration Architecture

■ **NEW (v1.1): Upgraded recommendation: use a VB.NET EXE wrapper instead of calling DLL directly from VBA. Direct VBA-to-DLL calls are unreliable with 64-bit Access.**

Architecture:

```

Access (VBA) ── Shell() ── VB.NET EXE ── Scanner ── (Hardware)
                  │          │          │          │
                  └─ (Biometric Wrap) ── (Hardware) ──
Exit Code ── Result
  
```

Workflow:

1. Access writes PupilID to a temp file (C:\Temp\psms_request.txt)
2. Access calls Shell("BiometricWrapper.exe enrol", vbHide)
3. VB.NET EXE: captures scan, processes template, writes result to temp file
4. Access reads result file: 1=success, 0=failure
5. Access updates ThumbprintData and ThumbprintEnrolled fields

11.3 Phase 1 Placeholder Functions

```

Function EnrolThumbprint(pupilID As Long) As Boolean
    ' TODO Phase 2: Replace with Shell() call to BiometricWrapper.exe
    MsgBox "Biometric enrolment not yet activated. Coming in Phase 2.", vbInformation
    EnrolThumbprint = False
End Function

Function VerifyThumbprint(pupilID As Long) As Boolean
    ' TODO Phase 2: Replace with Shell() call to BiometricWrapper.exe
    MsgBox "Biometric verification not yet activated. Coming in Phase 2.", vbInformation
    VerifyThumbprint = False
End Function
  
```

12. Testing Checklist

Run through every item before presenting to the school. Tick only when fully verified with real test data.

Database Structure

- All 13 tables created with correct fields and data types.
- All primary keys are AutoNumber.
- All foreign key fields are Number / Long Integer.
- tbl_PupilStatus pre-populated: 1=Active, 2=Left, 3=Graduated, 4=Transferred.
- tbl_AcademicYears has current year with IsCurrent = Yes.
- All 18 relationships built with referential integrity enforced.
- AdmissionNo in tbl_Pupils has Unique Index (No Duplicates).
- PhotoPath is Short Text (255), NOT OLE Object.

Forms

- frm_MainDashboard opens as startup form and shows live stats.
- frm_PupilRegistration: AdmissionNo auto-generates on new record.
- frm_PupilRegistration: Photo picker works and saves file path correctly.
- frm_ParentGuardian: linked guardians subform shows correctly.
- frm_Attendance: class filter loads correct pupils only.
- frm_Attendance: 'Mark All Present' stamps correct timestamp.
- frm_Attendance: 'Save' does NOT create duplicate records — test by saving twice.
- frm_Dismissal: exists, records DismissalTime correctly.
- frm_Dismissal: 'Mark All Dismissed' only updates Present pupils.
- frm_PickupLog: only CanPickUp = Yes guardians appear in list.
- frm_PickupLog: emergency PIN override logs OverridePINUsed = Yes.
- frm_FeePayment: overpayment is blocked with clear message.
- frm_FeePayment: ReceiptNumber generates in RCP-YYYY-NNNNN format.
- Academic Year dropdowns on all forms source from tbl_AcademicYears.

Queries & Reports

- qry_FeeSummary: test with pupil who has NO payments — should show zero paid, full balance.
- qry_FeeSummary: test with pupil who has PARTIAL payment — shows correct balance.

- qry_AttendanceSummary: percentages correct for test data.
- qry_ResultSummary: returns all subjects for a test pupil.
- rpt_ReportCard: renders cleanly for one pupil.
- rpt_FeeReceipt: correct receipt number, pupil name, amount, date.
- rpt_FeeDefaulters: shows only pupils with Balance > 0.
- rpt_EmergencyContacts: prints complete class-by-class contact list.
- All reports: school header, page numbers, and date visible.

VBA & Data Integrity

- GradeFromScore(39)='F', GradeFromScore(40)='D', GradeFromScore(70)='A' — test all boundaries.
- Duplicate AdmissionNo rejected at both VBA and database level.
- Deleting a class blocked if active pupils are linked.
- Deleting a pupil blocked if attendance or payment records exist.
- All required fields reject null on save attempt.
- CheckDatabaseSize() function runs on dashboard open without error.
- Photo picker saves file path correctly — image displays in form.

13. Document Changelog

Version 1.1 — Corrections by Engineer Review

Section	Change	Reason
2.3	Added Split Database requirement (MANDATORY)	Prevent multi-user corruption
2.5	Photo: OLE Object → PhotoPath (file path)	Prevent database bloat
3.3	tbl_Pupils: Photo field → PhotoPath (Short Text)	Same as above
4	Added 3 missing Staff relationships	Complete referential integrity
5.5	Added frm_Dismissal as required form	Workflow was incomplete
5.6	Added emergency override PIN to frm_PickupLog	Real-world safety requirement
6.1	qry_FeeSummary: INNER JOIN → LEFT JOIN	Include pupils with no payments
7.2	Save Attendance: INSERT/UPDATE logic added	Prevent duplicate records
9	Added rpt_EmergencyContacts	Operational necessity
10.7	Added PickPhoto() VBA function	Replaces OLE approach
10.8	Added CheckDatabaseSize() VBA function	Prevent hitting 2 GB limit
11.2	Upgraded to EXE wrapper recommendation	Stable biometric integration

Version 1.2 — Additional Fixes (This Document)

Section	Change	Reason
2.5	Added photo backup guidance	Prevent data loss
3.7	Added OverridePINUsed field to tbl_PickupLog	Audit trail for PIN overrides
3.8	AcademicYear → AcademicYearID (FK to lookup table)	Prevent typo mismatches
3.10	AcademicYear → AcademicYearID in tbl_ExamScores	Same as above
3.12	tbl_PupilStatus: added pre-population instructions	Prevent StatusID=1 confusion
3.13	tbl_AcademicYears: upgraded from Optional to Mandatory	Prevent free-text inconsistency
4	Added 3 more relationships (Status, AcademicYears)	Complete referential integrity
6.1	qry_FeeSummary uses AcademicYearID instead of text	Consistent with lookup table
10.2	ReceiptNumber: RCP-NNNNN → RCP-YYYY-NNNNN	Year-based audit trail
12	Checklist updated for all v1.2 changes	Accuracy

